Rafael Villagra de la Fuente

Advanced Aerospace Engineering Student
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LinkedIn
Engineering Portfolio

Education

•Politecnico di Torino & Universidad Nacional de Córdoba *09.2025 – Present*
Double Degree Program: Master's in Aerospace Engineering Turin, IT | Córdoba, AR

- Academic focus: Aerostructures and Aerogasdynamics.
- *Media Pesata dei Voti: 29.1*

•Universidad Nacional de Córdoba *03.2021 – Present*
Bachelor's in Aeronautical Engineering (Average: 7.03 - Eq. 25.4/30) Córdoba, AR

- Main courses: Fluid Mechanics, Structural Analysis I and II, Control Theory, Flight Mechanics.
- International Student Assistant & Student Tutor (2024 - 2025).

Professional Experience

•FDC Diseños y Desarrollos *12.2024 – 08.2025*
Engineering Assistant - Internship Córdoba, AR

- Full participation in the development of a UTV vehicle, including design and structural analysis.
- Experience in track engineering and telemetry data analysis in various projects.
- Preparation of technical design, structural analysis, and track performance reports.
- *Reference Contact:* Engineer Javier Ciabattari - jciabattari@fdceng.com

Relevant Academic Projects

•Conceptual Design of a Regional / Hybrid Aircraft *2025*
Aerodynamic and structural preliminary sizing

- Tools & technologies used: MATLAB, LaTeX, SolidWorks.
- Defined performance parameters, initial sizing, and feasibility study for a hybrid regional transport aircraft.

•Flight mechanics and performance of the Douglas DC-3 *2024*
Performance, static and dynamic stability analysis

- Determination of performance characteristics and presentation of data curves for various flight conditions.
- Developed within the **Flight Mechanics** course; *Reference:* jsaldia@unc.edu.ar

•Lift of a parabolic airfoil using the discrete vortex method *2024*
Computational aerodynamics

- Calculated airfoil lift, determined the approximate velocity field, and validated data through literature.
- Developed within the **Fluid Mechanics** course; *Reference:* andres.cimino@unc.edu.ar

Technical Skills & Certifications

•Engineering Software (CAD/CAE/CFD): Ansys, SolidWorks, CATIA, OpenFOAM

•Programming & Tools: MATLAB, Simulink, Python, LaTeX

•Languages: Spanish (Native), English (B2 - FCE 178/190), Italian (B1 - PLIDA 102/120)

•Certifications: Mastering ANSYS CFD (35h) | OpenFOAM for absolute beginners (5h) | Public Speaking